

Noise-Robust Classification of Ground Moving Targets Based on Time-Frequency Feature From Micro-Doppler Signature

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Abstract—A noise-robust classification method is proposed to discriminate the moving vehicle and walking human via the time-frequency feature extracted from the micro-Doppler signature of low resolution radar. Since the signal-to-noise ratio (SNR) directly relates to the distance between the target and radar for a given noise power and radar power, the denoising preprocessing is usually required to increase the classification distance between the target and radar in the real application. In this paper, we extend the real-valued probabilistic principal component analysis model to the complex-value domain, and develop the complex probabilistic principal component analysis (CPPCA) model for the complex-valued echoes from the ground moving targets. Then, the denoising preprocessing is accomplished based on signal reconstruction with CPPCA model, where we utilize the Bayesian inference criterion (BIC) to adaptively select the principal components. Compared with the existing CLEAN-based noise reduction method, the CPPCA-BIC-based method can work without SNR prior information. After denoising, a 3-D time-frequency feature vector is extracted from the denoised micro-Doppler signatures of the two kinds of ground targets, and the classification is performed via support vector machine classifier. In the experiments based on the measured data, the proposed classification scheme shows good classification and denoising performances under the relatively low SNR condition. The proposed method can also be applied to other classification problems based on micro-Doppler effect.

Index Terms—Micro-Doppler signature, radar target classification, time-frequency analysis, feature extraction, probabilistic principal component analysis (PPCA), Bayesian inference criterion (BIC), denoising preprocessing.

I. INTRODUCTION

WHEN radar illuminates a moving target, the carrier frequency of the returned signal will be shifted. This phenomenon is known as Doppler effect. If the target or any structural component of the target has oscillatory or rotation motion, which is referred to as micro-motion, such micro-motion may induce the additional Doppler modulation that

generates sideband frequencies around the target's Doppler frequency, which is called as micro-Doppler effect. In recent years, there have been lots of researches on applications of micro-Doppler effect in radar area [3], [6], [8], [16]. Papers [12], [17], [21], [22], [27] analyze the micro-Doppler effect in ISAR/SAR images. Low resolution radar imaging of rotating blades based on micro-Doppler effect is investigated in [24]. Papers [5], [9] show the micro-Doppler information contained in the time-frequency spectra of different human gait motions can be utilized for the human activity classification. A hierarchical micro-Doppler feature extraction method for discriminating the moving wheel and tracked vehicles is discussed in [11]. Papers [19], [20] study on the classification problem between pedestrians and vehicles. Some micro-Doppler feature extraction methods are proposed for aircraft targets classification in [4], [7], and [10], of which the recent paper [7] proposes a new feature extraction method based on Empirical Mode Decomposition (EMD) and CLEAN technique.

For radar target classification problem, the training data are usually collected via simulations or via some cooperative measurement experiments, where the signal-to-noise ratio (SNR) is usually high; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the non-cooperative targets and radar. As shown in the experimental results in [4] and [7], the classification performance dramatically deteriorates under the low SNR condition. Since the SNR directly relates to the distance between the target and radar for a given noise power and radar power, the noise robustness of a classification method is an important factor in increasing the classification distance between the target and radar in the real application. Based on the sparse representation theory, a simple noise reduction method is proposed in [7]. Nevertheless, this method requires the prior information about SNR. If the SNR is not estimated accurately, the classification performance will decrease.

The research reported here seeks an alternative approach to denoise the complex-valued low resolution radar echo based on Complex Probabilistic Principal Component Analysis (CPPCA) model. Then a 3-dimensional time-frequency feature vector is extracted from the denoised micro-Doppler signatures to distinguish the moving vehicle and walking human. In the real battlefield, since the vehicles and persons usually undertake the different tasks, it is important to discriminate these

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two kinds of targets [19], [20]. In the experiments based on measured data, the proposed method shows inspiring performances on classification and denoising for the test samples under low SNR. The proposed method can also be applied to other classification problems based on micro-Doppler effect.

Our contributions are as follows:

- *The Complex Probabilistic Principal Component Analysis (CPPCA) model is developed to model the complex-valued low-resolution radar echoes.* Principal Component Analysis (PCA) can construct the orthogonal signal and noise subspaces for a set of given data, and realize the noise reduction via reconstructing the data within the signal subspace and discarding the noise subspace. Therefore, under such a formulation, the precise estimation of the dimensionality of signal subspace is the central issue. There are some threshold-based schemes, e.g., setting a threshold for the ratio of each eigenvalue to the total power, proposed to separate the eigenvectors into signal and noise subspaces [25]. However, these methods require a user defined threshold and their dimensionality estimation results are not precise in most cases. Paper [23] introduces the probabilistic model into the PCA framework via modeling the signal and noise components with the independent Gaussian distributions, which is referred to as Probabilistic Principal Component Analysis (PPCA) model. Under such probabilistic formulation, some adaptive model selection methods can be applied to dimensionality estimation for the signal subspace, and the estimation precision can be improved. Since the radar echoes can be regarded as complex data consisting of the inphase and quadrature components, we extend the real-valued PPCA model in [23] to the complex-value domain, and develop the Complex Probabilistic Principal Component Analysis (CPPCA) model for the complex-valued low-resolution radar echoes.
- *A new noise reduction method for low resolution radar echo is proposed based on signal reconstruction with CPPCA model, where we utilize the Bayesian Inference Criterion (BIC) to adaptively select the principal components.* The Akaike's Information Criterion (AIC) [1] and BIC [18] are two classical model selection methods, which are commonly used in time series and regression analysis. As discussed in [14], AIC is not consistent and tends to overestimate the dimensionality of signal subspace for the large sample size; while BIC is consistent and estimates more precisely for the large sample size. Here we select BIC, and derive the model selection formula based on BIC for selecting the complex-valued principal components under the CPPCA formulation. Then the denoising preprocessing is accomplished via reconstructing the echo within the BIC-determined signal subspace and discarding the residual noise subspace. Compared with the existing CLEAN-based noise reduction method [7], the CPPCA-BIC-based method can work without SNR prior information.
- *A low-dimensional feature vector is extracted from the denoised micro-Doppler signatures to distinguish the moving vehicle and walking human.* After making an

analysis on the time-frequency spectrograms of the measured data, we find differences between the two kinds of ground moving targets, and propose a three-dimensional feature vector to describe such discriminative information.

The reminder of the paper is organized as follows. Section II makes an analysis on the measured data of the moving vehicle and walking human, and proposes the feature extraction method from time-frequency spectrogram. To deal with the low SNR problem of the test data measured from the far distance, the noise reduction method based on CPPCA-BIC is developed, and the noise-robust classification scheme is introduced in Section III. The classification results based on the measured data is given in Section IV, followed by the conclusion in Section V.

II. FEATURE EXTRACTION FROM TIME-FREQUENCY SPECTROGRAM OF MICRO-DOPPLER SIGNATURE

A. Measurement of Moving Vehicle and Walking Human

The measured data of the walking human and moving vehicle are collected with a low resolution radar, which works at Ka waveband and under the high SNR condition ($\text{SNR} \approx 35\text{dB}$). Single person or tracked vehicle moves away or towards radar in the measurement experiment. Since the time-frequency analysis can show how the instantaneous frequency components of a signal vary with time, it is a good tool to analyze the micro-Doppler characteristic of a target signal. Figure 1 shows the time-frequency spectrograms of some measured echoes via the short time Fourier transform (STFT). The torso-motion, micro-motion and clutter components are indicated in Figures 1(a) and (b) for the data of a walking human; while the bulk-motion, micro-motion, upper-track micro-motion and clutter components are indicated in Figures 1(c) and (d) for the data of a moving vehicle. As shown in Figure 1(c), the upper-track micro-motion can induce the $2v$ -Doppler component, where v denotes the velocity of bulk-motion. Nevertheless, it should be pointed out that the $2v$ -Doppler component can not be always observed due to the shelter of upper-track in practice, as shown in Figure 1(d). [11] makes a detailed analysis on the micro-motion model for a moving tracked vehicle.

Since the arms and legs swing, the micro-motion components are generated in the echo from a walking human. In theory, the sub-echoes of swinging arms or legs are time-varying functions as $A_0 \sin(\omega_0 t + \theta_0)$ with A_0 denoting the swing amplitude, ω_0 denoting the swing frequency and θ_0 denoting the initial swing phase. On the contrary, the micro-Doppler components induced by the rolling of track are not time-varying, and are within the Doppler frequency range between 0 and $2v/\lambda$ with v denoting the velocity of bulk-motion and λ denoting the wavelength. In accordance with the above analysis, the instantaneous micro-Doppler components of a walking human vary with the time periodically in Figures 1(a) and (b); while the instantaneous micro-Doppler components of a moving vehicle do not show such periodical property in Figures 1(c) and (d). In addition, the energy of micro-motion components of the moving vehicle is much smaller than that

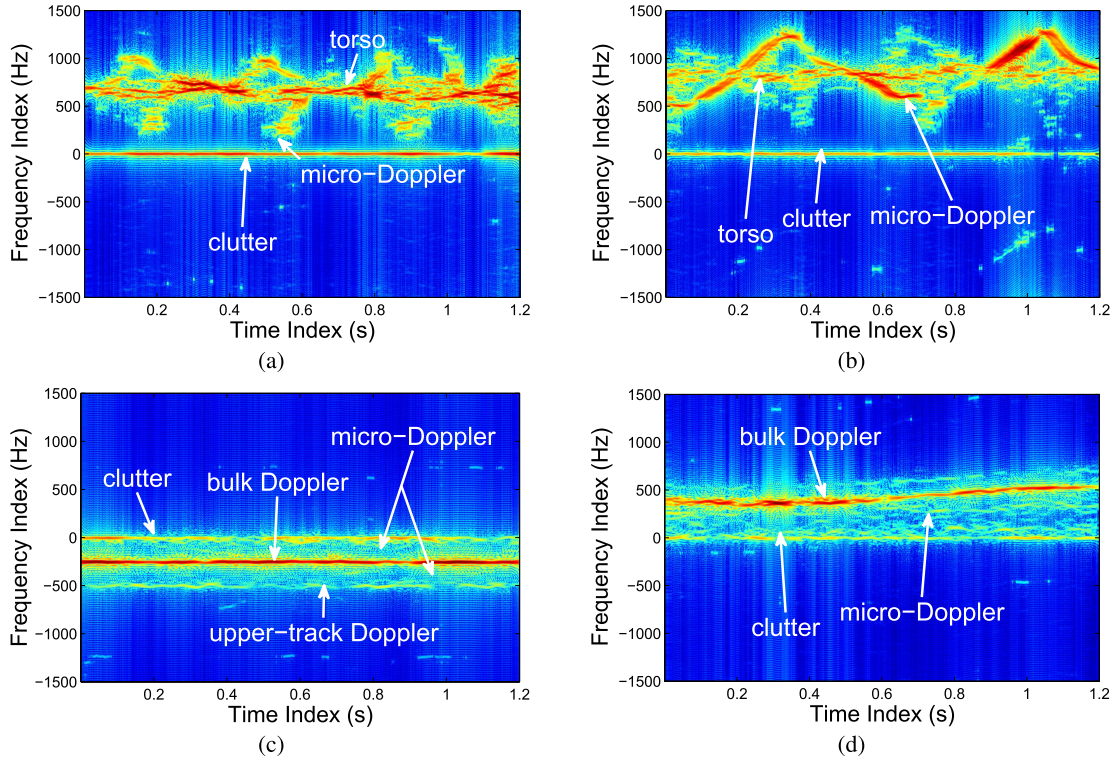


Fig. 1. Time-frequency spectrograms of some measured echoes under the high SNR condition ($\text{SNR} \approx 35\text{dB}$): (a) walking human; (b) walking human; (c) moving vehicle; and (d) moving vehicle.

of bulk-motion component; while this difference between the energies of micro-motion and torso-motion components of the walking human is not so evident.

B. Time-Frequency Feature Extraction

As shown in Figure 1, we can find some differences in the time-frequency spectrograms between the two kinds of ground moving targets. To avoid the influence from the clutter component on the following feature extraction, we first use the CLEAN algorithm to remove the ground clutter in the measured data. CLEAN is a classical signal deconvolution technique originally for improving the quality of microwave imaging [26] that can successively remove the large harmonic components from a signal. Paper [19] proposes to utilize the CLEAN technique for the ground clutter suppression. Then we use the following three features to depict the distinctions between the two targets.

- Feature 1: $\mathcal{F}(1) = \text{entropy}(S(t, f)) = -\sum_{t=1}^T \sum_{f=1}^F p_{t,f} \log(p_{t,f})$ with $p_{t,f} = \frac{S(t,f)}{\sum_{t'=1}^T \sum_{f'=1}^F S(t',f')}$, which represents the entropy of the time-frequency spectrogram $S(t, f)$. The time-frequency spectrogram of a signal shows the energy distributions of all instantaneous frequency components. The more the energy gathers, which means a spectrogram is close to an impulse function, the smaller the entropy is. As shown in Figure 1, the spectrogram of a moving vehicle gathers closer than that of a walking human, since the energy of bulk-motion component is much larger than that of micro-motion component for

TABLE I
FEATURE VALUES OF EACH SUBFIGURE IN FIGURE 1, WHERE THE GROUND CLUTTER COMPONENT IN THE MEASURED DATA IS REMOVED VIA THE CLEAN ALGORITHM [19]

	Figure 1 (a)	Figure 1 (b)	Figure 1 (c)	Figure 1 (d)
$\mathcal{F}(1)$	7.28	7.07	6.70	6.86
$\mathcal{F}(2)$	3.45	3.03	1.61	1.85
$\mathcal{F}(3)$	0.49	0.59	0.80	0.78

a moving vehicle. Therefore, as demonstrated in Table I, the entropy of the time-frequency spectrogram of a walking human is larger than that of a moving vehicle.

- Feature 2: $\mathcal{F}(2) = \text{entropy}\left(\frac{G(f)}{\sum_{f'=1}^F G(f')}\right) = -\sum_{f=1}^F \rho_f \log(\rho_f)$ with $\rho_f = \frac{G(f)}{\sum_{f'=1}^F G(f')}$ and $G(f) = \text{FFT}(g(t)) = \text{FFT}\left(\frac{\sum_{f=1}^F f \cdot |S(t,f)|^2}{\sum_{f=1}^F |S(t,f)|^2}\right)$, which represents the entropy of the average Fourier spectrum $G(f)$. For a given t_0 , the time-frequency spectrogram $S(t_0, f)$ shows the instantaneous frequencies of all scatterers on the target at time t_0 , then $g(t) = \frac{\sum_{f=1}^F f \cdot |S(t,f)|^2}{\sum_{f=1}^F |S(t,f)|^2}$ can represent the average instantaneous frequency of the target signal. Since $g(t)$ is a linear combination of the instantaneous frequencies of all scatterers, all of the instantaneous frequency components within the time interval $[1, T]$ can be got via Fourier transform on $g(t)$, i.e., the average Fourier spectrum $G(f) = \text{FFT}(g(t))$ with $\text{FFT}(\cdot)$ denoting the fast Fourier transform. As shown in Figure 1, since the

instantaneous micro-Doppler components of a walking human show periodical property, its average Fourier spectrum should contain the corresponding periodical frequency components; while the frequency components in the average Fourier spectrum of a moving vehicle should be much less, since its instantaneous frequency components do not periodically vary with the time. Therefore, as demonstrated in Table I, the entropy of the average Fourier spectrum of a walking human is larger than that of a moving vehicle.

- Feature 3: $\mathcal{F}(3) = \max(\rho)$ represents the maximum value of the normalized average Fourier spectrum $\rho = \{\rho_f\}_{f=1}^F$. The more the energy of ρ gathers, the larger the maximum value is. As shown in Figure 1, the instantaneous frequency component corresponding to the bulk-motion takes up a large portion of the whole energy in the time-frequency spectrogram of a moving vehicle. Therefore, as demonstrated in Table I, the maximum value of the normalized average Fourier spectrum of a moving vehicle is larger than that of a walking human.

Figure 2 shows the 3-dimensional dispersion situations of the above three features from three different view-aspects for some measured data under the high SNR condition (SNR $\approx$ 35dB), where the green circle markers denote the feature data of a moving vehicle, the black star markers denote the feature data of a walking human and the blue plus markers denote the Support Vector Machine (SVM) [2] classifier boundary learned via the feature data shown in this figure. Before putting the feature data into a classifier, we should first normalize the feature data via subtracting their mean and divided by their standard deviation, which is a common preprocessing method to unify the scales of different features before classification. Here Figure 2 shows the normalized feature data, therefore, their values are quite different from the original feature values shown in Table I and Figure 6. We can see these features can depict the distinctions between the two kinds of ground moving targets. The average classification accuracy rate of the feature data via the boundary in Figure 2 is about 97%.¹

III. NOISE-ROBUST CLASSIFICATION METHOD BASED ON COMPLEX PROBABILISTIC PRINCIPAL COMPONENT ANALYSIS

In Section II-B, we consider the classification problem of moving vehicle and walking human under the high SNR condition. In the real application of radar target classification, the training data are usually collected via simulations or via some cooperative measurement experiments, where the signal-to-noise ratio (SNR) is usually high; while the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to

the severe measurement conditions, such as the large distance between the non-cooperative targets and radar. According to our experience with a large amount of measured data, the SNR of measured low-resolution radar data from some non-cooperative measurement experiments is usually ≤ 10 dB. As shown in the experimental results in [4] and [7], the classification performance dramatically deteriorates under the low SNR condition. Since the SNR directly relates to the distance between the target and radar for a given noise power and radar power, the noise robustness of a classification method is an important factor in increasing the classification distance between the target and radar in the real application. In this section, we propose a noise reduction preprocessing approach to denoise the measured test data and further to improve the noise-robustness of the classification method.

A. Noise Reduction Preprocessing Approach

The PCA model can construct the orthogonal signal and noise subspaces for a set of given data, and realize the noise reduction via reconstructing the data within the signal subspace and discarding the noise subspace. The PPCA [23] develops a probabilistic model for not only the signal but also the noise components. An obvious advantage of PPCA over PCA is that some adaptive model selection methods can be applied for data denoising under the probabilistic formulation. The conventional PPCA model is developed in the real-value domain. Since the radar target echoes include the inphase and quadrature components, which are complex-valued data, we extend the real-valued PPCA model in [23] to the complex-value domain, and develop the CPPCA model for the complex-valued low-resolution radar echoes to accomplish denoising.

Given a received low-resolution echo $\mathbf{x} \in \mathbb{C}^{M \times 1}$, it can be modeled via CPPCA as

$$\mathbf{x} = \mathbf{W}\mathbf{y} + \boldsymbol{\mu} + \boldsymbol{\epsilon} \quad (1)$$

In (1), $\mathbf{W} \in \mathbb{C}^{M \times d}$ with $d < M$ denotes the orthogonal projection matrix with $\mathbf{w}_p^H \mathbf{w}_q = c$ ($p, q = 1, 2, \dots, d$), where $c \neq 0$ for $p = q$ and $c = 0$ for $p \neq q$, i.e., $\mathbf{W}^H \mathbf{W}$ is a diagonal matrix; $\mathbf{y} \in \mathbb{C}^{d \times 1}$ denotes the latent variables, and $\mathbf{y} \sim \text{CNorm}(0, \mathbf{I}_d)$ with $\text{CNorm}(\cdot)$ representing the complex Gaussian distribution with $p(\mathbf{y}|\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = \frac{1}{(\pi)^d |\boldsymbol{\Sigma}_0|} \exp\{-(\mathbf{y} - \boldsymbol{\mu}_0)^H \boldsymbol{\Sigma}_0^{-1} (\mathbf{y} - \boldsymbol{\mu}_0)\}$; $\boldsymbol{\mu} \in \mathbb{C}^{M \times 1}$ denotes the mean vector of the observed data; $\boldsymbol{\epsilon} \in \mathbb{C}^{M \times 1}$ denotes the noise component, which is independent from the signal component $\mathbf{y}$, and $\boldsymbol{\epsilon} \sim \text{CNorm}(0, \sigma^2 \mathbf{I}_M)$. Then the radar echo $\mathbf{x}$ should follow the complex Gaussian distribution, i.e., $\mathbf{x} \sim \text{CNorm}(\boldsymbol{\mu}, \mathbf{C} = \mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M)$. The corresponding density is

$$\begin{aligned} p(\mathbf{x}) &= \frac{1}{(\pi)^M |\mathbf{C}|} \exp\left[-(\mathbf{x} - \boldsymbol{\mu})^H \mathbf{C}^{-1} (\mathbf{x} - \boldsymbol{\mu})\right] \\ &= \frac{1}{(\pi)^M |\sigma^2 \mathbf{I}_M + \mathbf{W}\mathbf{W}^H|} \\ &\quad \cdot \exp\left[-(\mathbf{x} - \boldsymbol{\mu})^H (\sigma^2 \mathbf{I}_M + \mathbf{W}\mathbf{W}^H)^{-1} (\mathbf{x} - \boldsymbol{\mu})\right] \end{aligned} \quad (2)$$

¹In order to present the 3D dispersion situation of the three features more clearly, an animation with more views is provided as the supplementary material of this paper. In this animation, we also give the 3D views of the features from the measured data under SNR = 10dB and those of the denoised features via CPPCA-BIC for comparison, which respectively correspond to Figures 7 and 8.

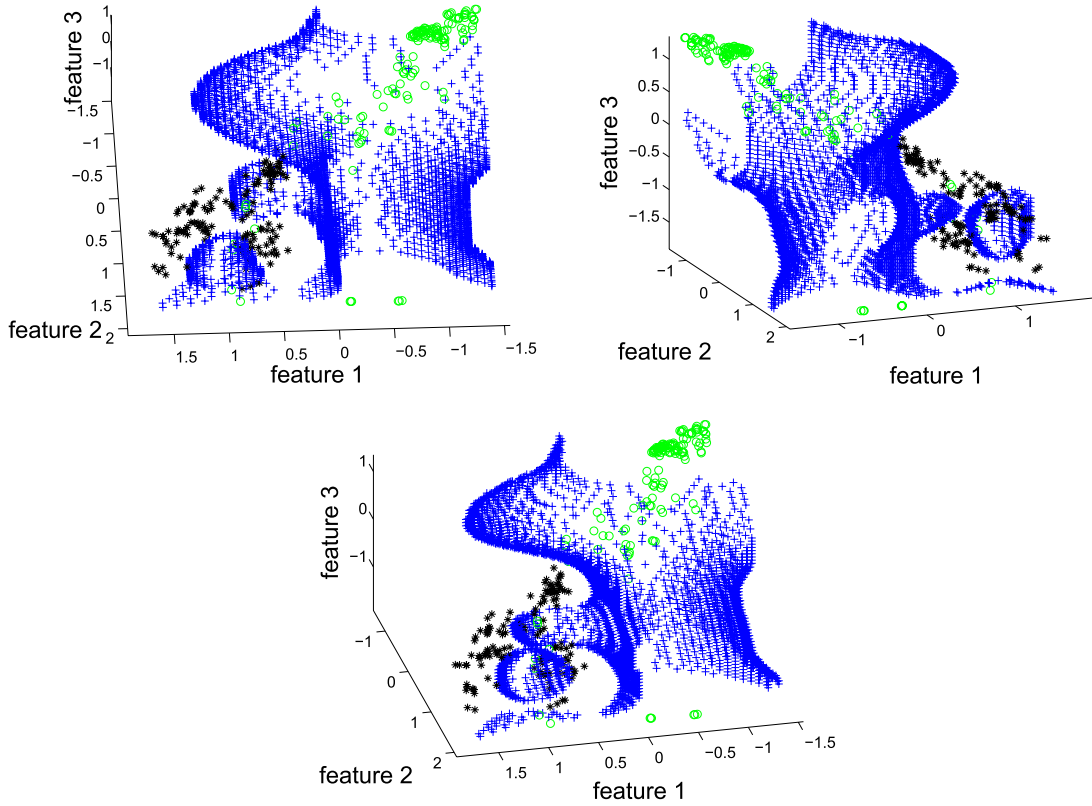


Fig. 2. Three-dimensional dispersion situations of the proposed time-frequency features from three different view-aspects for some measured data under the high SNR condition (SNR $\approx$ 35dB). The green circle markers denote the feature data of a moving vehicle, the black star markers denote the feature data of a walking human and the blue plus markers denote the SVM classifier boundary learned via the feature data shown in this figure. The average classification accuracy rate of the feature data via the boundary in this figure is about 97%. The feature data shown in this figure are normalized with their mean and standard deviation.

The complex echo from a range cell is the summation of the complex-valued scattering responses from all scatterers in this range cell. According to the central limit theorem, the summation of many independent variables approximately follows the Gaussian distribution. Therefore, for the sake of simplicity, it is usually reasonable to assume that the real and imaginary parts of the complex echo from a range cell respectively follow the Gaussian distributions, and the complex-valued variable follows the complex Gaussian distribution.

Based on (1), the log-likelihood function for a set of observed data $\{\mathbf{x}_n\}_{n=1}^N$ is

$$\begin{aligned} \ell(\Phi) &= \sum_{n=1}^N \log(p(\mathbf{x}_n)) \\ &= -N \left(\log|\mathbf{C}| + \text{Tr}(\mathbf{C}^{-1}\mathbf{S}) + M \log \pi \right) \\ &= -N \left(\log|\sigma^2 \mathbf{I}_M + \mathbf{W}\mathbf{W}^H| \right. \\ &\quad \left. + \text{Tr}((\sigma^2 \mathbf{I}_M + \mathbf{W}\mathbf{W}^H)^{-1}\mathbf{S}) + M \log \pi \right) \end{aligned} \quad (3)$$

with

$$\mathbf{S} = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^H \quad (4)$$

where $\Phi = \{\boldsymbol{\mu}, \sigma^2, \mathbf{W}\}$ denotes the model parameters, N is the number of measurements in the data set, $\text{Tr}(\cdot)$ represents

to get the trace of the matrix inside parentheses, $\mathbf{S}$ denotes the covariance matrix of measurements.

To get the maximum likelihood estimator for $\mathbf{W}$, we let the derivation of the log-likelihood with respect to $\mathbf{W}^*$, i.e., the conjugate complex derivative, be a zero matrix.

$$\frac{\partial \ell}{\partial \mathbf{W}^*} = -N \left(\frac{\partial \log|\mathbf{C}|}{\partial \mathbf{W}^*} + \frac{\partial \text{Tr}(\mathbf{C}^{-1}\mathbf{S})}{\partial \mathbf{W}^*} \right) = \mathbf{0} \quad (5)$$

According to (179) in [15],

$$\frac{\partial \log|\mathbf{C}|}{\partial \mathbf{W}^*} = \frac{\text{Tr}((\frac{\partial \log|\mathbf{C}|}{\partial \mathbf{C}})^T \partial \mathbf{C})}{\partial \mathbf{W}^*} + \frac{\text{Tr}((\frac{\partial \log|\mathbf{C}|}{\partial \mathbf{C}^*})^T \partial \mathbf{C}^*)}{\partial \mathbf{W}^*} \quad (6)$$

Based on (37) and (173) in [15],

$$\begin{aligned} \frac{\partial \log|\mathbf{C}|}{\partial \mathbf{C}} &= \frac{\text{Tr}(\mathbf{C}^{-1} \partial \mathbf{C})}{\partial \mathbf{C}} \\ &= \frac{1}{2} \left(\frac{\text{Tr}(\mathbf{C}^{-1} \partial \mathbf{C})}{\partial \text{Real}(\mathbf{C})} - i \frac{\text{Tr}(\mathbf{C}^{-1} \partial \mathbf{C})}{\partial \text{Imag}(\mathbf{C})} \right) \\ &= \frac{1}{2} (\mathbf{C}^{-1} + \mathbf{C}^{-1}) = \mathbf{C}^{-1} \end{aligned} \quad (7)$$

where $\text{Real}(\cdot)$ denotes the real part of the matrix inside parentheses, and $\text{Imag}(\cdot)$ denotes the imaginary part of the matrix inside parentheses. Similarly,

$$\frac{\partial \log|\mathbf{C}|}{\partial \mathbf{C}^*} = \frac{1}{2} (\mathbf{C}^{-1} - \mathbf{C}^{-1}) = \mathbf{0} \quad (8)$$

Substitute (7) and (8) into (6), we can get

$$\frac{\partial \log |C|}{\partial \mathbf{W}^*} = \frac{\text{Tr}(\mathbf{C}^{-1} \partial \mathbf{C})}{\partial \mathbf{W}^*} \quad (9)$$

Since $\mathbf{C} = \mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M$, then

$$\begin{aligned} \frac{\text{Tr}(\mathbf{C}^{-1} \partial \mathbf{C})}{\partial \mathbf{W}^*} &= \frac{\text{Tr}(\mathbf{W}^H \mathbf{C}^{-1} \partial \mathbf{W} + \mathbf{C}^{-1} \mathbf{W} \partial \mathbf{W}^H)}{\partial \mathbf{W}^*} \\ &= \frac{1}{2} \left(\frac{\text{Tr}(\mathbf{W}^H \mathbf{C}^{-1} \partial \mathbf{W} + \mathbf{C}^{-1} \mathbf{W} \partial \mathbf{W}^H)}{\partial \text{Real}(\mathbf{W})} + i \frac{\text{Tr}(\mathbf{W}^H \mathbf{C}^{-1} \partial \mathbf{W} + \mathbf{C}^{-1} \mathbf{W} \partial \mathbf{W}^H)}{\partial \text{Imag}(\mathbf{W})} \right) \\ &= \mathbf{C}^{-1} \mathbf{W} \end{aligned} \quad (10)$$

Therefore, $\frac{\partial \log |C|}{\partial \mathbf{W}^*} = \mathbf{C}^{-1} \mathbf{W}$. We can also get $\frac{\partial \text{Tr}(\mathbf{C}^{-1} \mathbf{S})}{\partial \mathbf{W}^*} = -\mathbf{C}^{-1} \mathbf{S} \mathbf{C}^{-1} \mathbf{W}$ via the similar derivation. Thus

$$\frac{\partial \log L}{\partial \mathbf{W}^*} = -N(\mathbf{C}^{-1} \mathbf{S} \mathbf{C}^{-1} \mathbf{W} - \mathbf{C}^{-1} \mathbf{W}) = \mathbf{0} \quad (11)$$

Since $\mathbf{W} \neq \mathbf{0}$, if $\mathbf{C} = \mathbf{S}$, we can substitute $\mathbf{S} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$, where the matrixes $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M]$ and $\mathbf{\Lambda} = \text{Diag}(\lambda_1, \lambda_2, \dots, \lambda_M)$ are respectively composed of the eigenvectors $\{\mathbf{u}_q\}_{q=1}^M$ and eigenvalues $\{\lambda_q\}_{q=1}^M$ of $\mathbf{S}$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_M$, and $\mathbf{C} = \mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M$, then

$$\mathbf{W}\mathbf{W}^H = \mathbf{U}_d \mathbf{\Lambda}_d \mathbf{U}_d^H - \sigma^2 \mathbf{I}_M = \mathbf{U}_d (\mathbf{\Lambda}_d - \sigma^2) \mathbf{U}_d^H \quad (12)$$

where $\mathbf{U}_d = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d]$ and $\mathbf{\Lambda}_d = \text{Diag}(\lambda_1, \lambda_2, \dots, \lambda_d)$. Thus the maximum likelihood estimator of $\mathbf{W}$ is

$$\mathbf{W}_{\text{ML}} = \mathbf{U}_d (\mathbf{\Lambda}_d - \sigma^2)^{1/2} \mathbf{T} \quad (13)$$

where $\mathbf{T}$ is an arbitrary rotation matrix, and can be selected as an identity matrix $\mathbf{I}_d$ in the real application for the sake of simplicity. If $\mathbf{C} \neq \mathbf{S}$,

$$\begin{aligned} \mathbf{S} \mathbf{C}^{-1} \mathbf{W} &= \mathbf{S} (\mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M)^{-1} \mathbf{W} \\ &= \mathbf{S} \mathbf{W} (\mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M)^{-1} \end{aligned} \quad (14)$$

Let $\mathbf{W} = \mathbf{U} \mathbf{L} \mathbf{V}^H$, where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M]$ and $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_d]$ are unitary matrices and $\mathbf{L} = \text{Diag}(l_1, l_2, \dots, l_d, 0, \dots, 0)$ is composed of the singular values of $\mathbf{W}$ with $l_1 \geq l_2 \geq \dots \geq l_d$.

$$\begin{aligned} \mathbf{S} \mathbf{C}^{-1} \mathbf{W} &= \mathbf{S} \mathbf{W} (\mathbf{W}\mathbf{W}^H + \sigma^2 \mathbf{I}_M)^{-1} \\ &= \mathbf{S} \mathbf{U} \mathbf{L} (\mathbf{L}^2 + \sigma^2 \mathbf{I}_M)^{-1} \mathbf{V}^H \end{aligned} \quad (15)$$

Since $\mathbf{C}^{-1} \mathbf{S} \mathbf{C}^{-1} \mathbf{W} = \mathbf{C}^{-1} \mathbf{W}$,

$$\mathbf{S} \mathbf{U} \mathbf{L} (\mathbf{L}^2 + \sigma^2 \mathbf{I}_M)^{-1} \mathbf{V}^H = \mathbf{U} \mathbf{L} \mathbf{V}^H \quad (16)$$

Therefore, $\mathbf{S} \mathbf{U} = \mathbf{U} (\mathbf{L}^2 + \sigma^2 \mathbf{I}_M)$. Then $\mathbf{S} = \mathbf{U} (\mathbf{L}^2 + \sigma^2 \mathbf{I}_M) \mathbf{U}^H = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^H$, where the matrixes $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_M]$ and $\mathbf{\Lambda} = \text{Diag}(\lambda_1, \lambda_2, \dots, \lambda_M) = \text{Diag}(l_1^2 + \sigma^2, l_2^2 + \sigma^2, \dots, l_d^2 + \sigma^2, \sigma^2, \dots, \sigma^2)$ are respectively composed of the eigenvectors $\{\mathbf{u}_q\}_{q=1}^M$ and eigenvalues $\{\lambda_q\}_{q=1}^M$ of $\mathbf{S}$ with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_M$. Thus we can also get $\mathbf{W}_{\text{ML}} = \mathbf{U}_d (\mathbf{\Lambda}_d - \sigma^2)^{1/2} \mathbf{T}$ with $\mathbf{U}_d = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d]$, $\mathbf{\Lambda}_d = \text{Diag}(\lambda_1, \lambda_2, \dots, \lambda_d)$ and $\mathbf{T}$ being an arbitrary rotation matrix.

We can easily get the maximum likelihood estimators for the other two parameters $\boldsymbol{\mu}$ and σ^2 as

$$\begin{aligned} \boldsymbol{\mu}_{\text{ML}} &= \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n \\ \sigma_{\text{ML}}^2 &= \frac{1}{M-d} \sum_{q=d+1}^M \lambda_q \end{aligned} \quad (17)$$

As shown in (13) and (17), there is still a parameter to be determined, *i.e.*, d . Unlike AIC, BIC is a consistent model selection method, which can achieve a better estimation result for the large sample size [14]. Paper [13] applies the BIC to the real-valued PPCA model to estimate d . Since the radar echoes considered in this paper are complex, here we derive the model selection formula based on BIC for the CPPCA model. The general formula of BIC is defined as [18]

$$\begin{aligned} d^* &= \arg \min \text{BIC}(d) \\ \text{BIC}(d) &= -2\ell(\boldsymbol{\Phi}) + \ln N \cdot Q(d) \end{aligned} \quad (18)$$

where $\boldsymbol{\Phi}$ denotes the model parameters, $\ell(\boldsymbol{\Phi})$ denotes the log-likelihood of the observed data set, $Q(d)$ denotes the freedom of the model parameters. For our CPPCA model, $\boldsymbol{\Phi} = \{\boldsymbol{\mu}, \sigma^2, \mathbf{W}\}$, $\ell(\boldsymbol{\Phi})$ has been derived in (3), the freedom of $\boldsymbol{\mu} \in \mathbb{C}^{M \times 1}$ is $2M$, the freedom of σ^2 is 1, and the freedom of $\mathbf{W} = \mathbf{U}_d (\mathbf{\Lambda}_d - \sigma^2)^{1/2} \mathbf{I}_d \in \mathbb{C}^{M \times d}$ with $\mathbf{U}_d^H \mathbf{U}_d = \mathbf{I}_d$ is $2Md - d(d-1) + 1$, then $Q(d) = 2(Md - \frac{d(d-1)}{2}) + M + 1$. Thus the BIC formula for our CPPCA model with ignoring the terms independent on d is

$$d^* = \arg \min \left(\frac{2N \log |\mathbf{C}_{\text{ML}}| + 2N \text{Tr}(\mathbf{C}_{\text{ML}}^{-1} \mathbf{S})}{+2 \ln N \cdot (Md - \frac{d(d-1)}{2})} \right) \quad (19)$$

where $\mathbf{C}_{\text{ML}} = \sigma_{\text{ML}}^2 \mathbf{I}_M + \mathbf{W}_{\text{ML}} \mathbf{W}_{\text{ML}}^H$ is dependent on d .

After all the parameters are estimated, the noise reduction for a noised echo $\mathbf{x}_n$ can be achieved via reconstructing the echo within the BIC-determined signal subspace and discarding the residual noise subspace.

$$\tilde{\mathbf{x}}_n = \mathbf{U}_{d^*} \mathbf{U}_{d^*}^H \mathbf{x}_n \quad (20)$$

where $\tilde{\mathbf{x}}_n$ denotes the denoised echo from $\mathbf{x}_n$, $\mathbf{U}_{d^*} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_{d^*}]$. As discussed above, the CPPCA-BIC-based noise reduction method doesn't require the noise power or SNR prior information during the denoising procedure.

B. Noise-Robust Classification Scheme

A typical noise-robust classification scheme is depicted in Figure 3. There are two stages, *i.e.*, training stage and classification stage, in the classification procedure.

In the training stage, the receiver noise component can be ignored, since the training data are collected via the cooperative measurement experiment, where the SNR is high. Firstly, we use the CLEAN algorithm to remove the ground clutter in the training data. Secondly, the three features discussed in Section II-B are extracted for each training sample. Finally, the normalized feature vectors $\tilde{\mathcal{F}} = [\tilde{\mathcal{F}}(1), \tilde{\mathcal{F}}(2), \tilde{\mathcal{F}}(3)]$ from all training samples are used to train the SVM classifier [2].

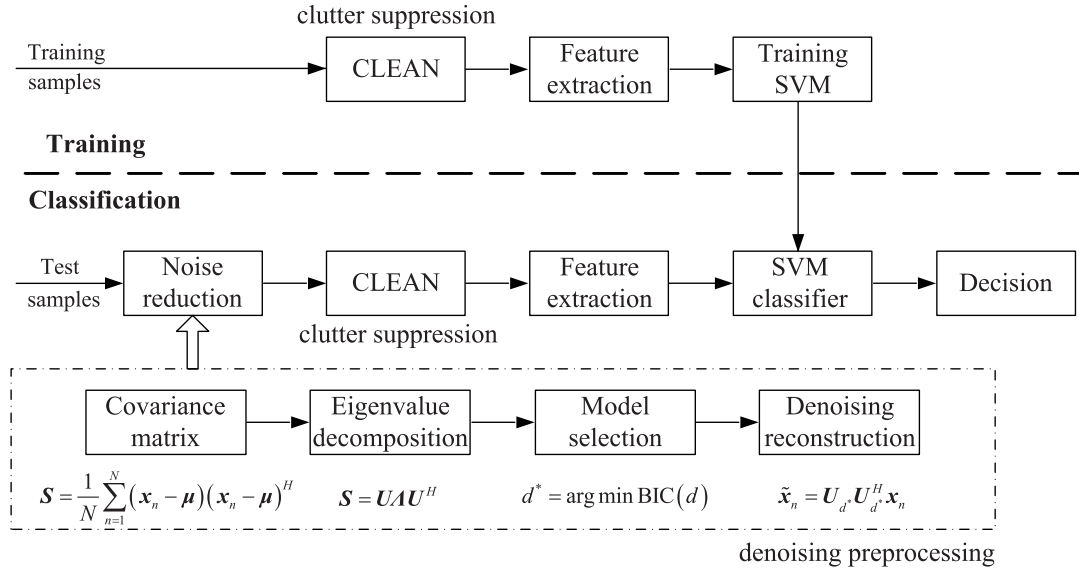
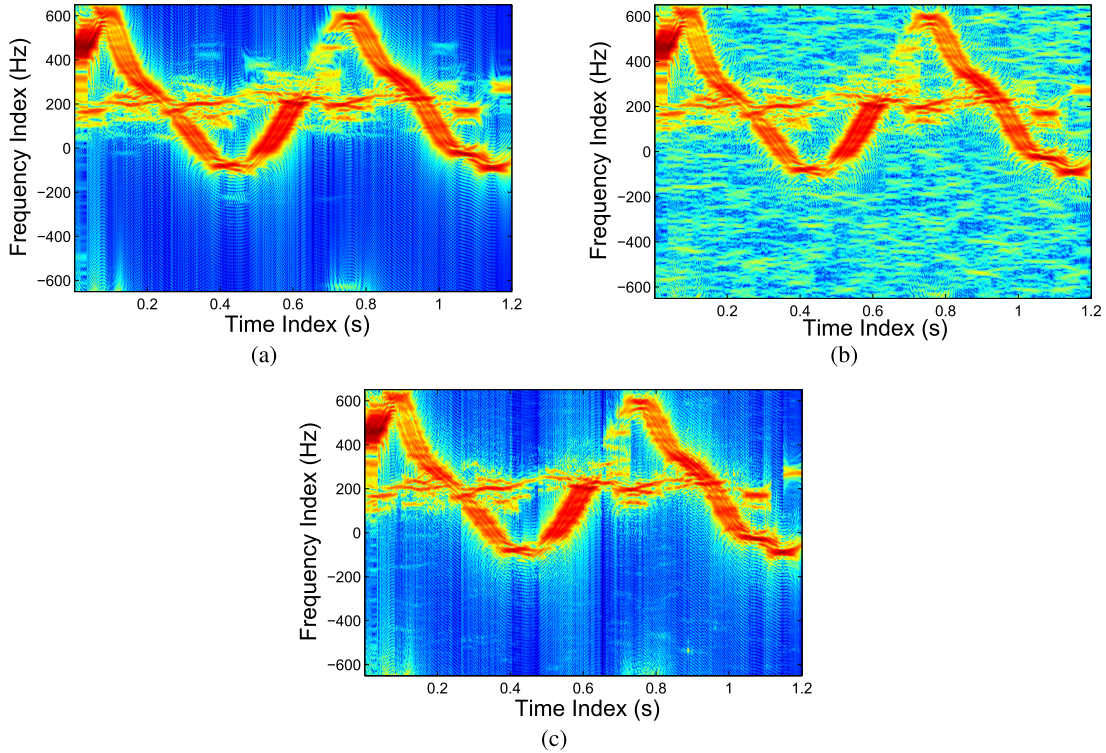


Fig. 3. A flow chart of our classification method.

Fig. 4. Time-frequency spectrograms of the measured echoes from a walking human: (a) SNR $\approx$ 35dB; (b) SNR = 10dB; and (c) denoised result from (b) via CPPCA-BIC.

In the classification stage, the test samples are usually got in the non-cooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the non-cooperative targets and radar. For a given noise power and radar power, the larger the distance between the target and radar is, the lower the SNR of the measured data is. Therefore we use the CPPCA-BIC-based noise reduction preprocessing approach, which are introduced in Section III-A, to denoise each test sample. Then the ground clutter component is removed, and the normalized feature

vector $\tilde{\mathcal{F}} = [\tilde{\mathcal{F}}(1), \tilde{\mathcal{F}}(2), \tilde{\mathcal{F}}(3)]$ is extracted from each denoised test sample. Finally, the SVM classifier learned in the training stage is used to give the classification decision for the feature vector of each test sample.

IV. EXPERIMENTAL RESULTS

A. Performance on Noise Reduction

Figures 4 and 5 respectively show the noise reduction performance for the measured data from a walking human and a moving vehicle. The time-frequency spectrum of the walking

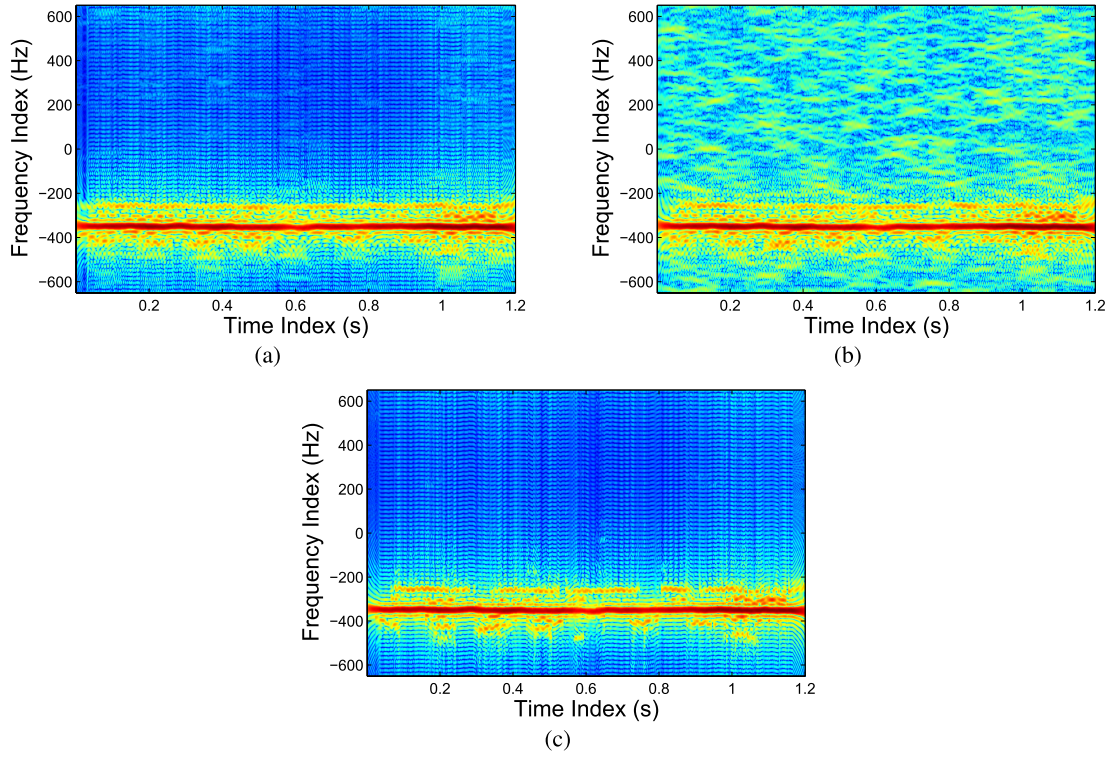


Fig. 5. Time-frequency spectrograms of the measured echoes from a moving vehicle: (a) $\text{SNR} \approx 35\text{dB}$; (b) $\text{SNR} = 10\text{dB}$; and (c) denoised result from (b) via CPPCA-BIC.

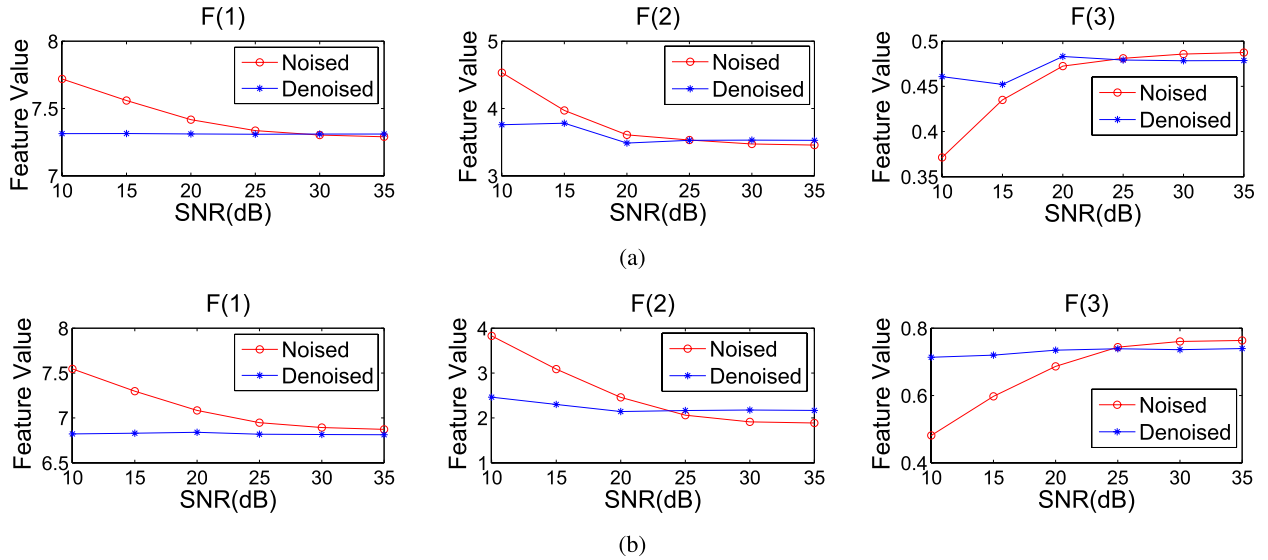


Fig. 6. Variation of the feature values from the measured data in Figures 4 and 5 with SNR. The red lines denote the feature values from the noised data, while the blue lines denote the feature values from the data denoised via CPPCA-BIC. (a) $\mathcal{F}(1)$, $\mathcal{F}(2)$ and $\mathcal{F}(3)$ from the measured data in Figure 4. (b) $\mathcal{F}(1)$, $\mathcal{F}(2)$ and $\mathcal{F}(3)$ from the measured data in Figure 5.

human in Figure 4(a) is under the high SNR condition of about 35dB and that in Figure 4(b) is under the SNR condition of 10dB. Figure 4(c) depicts the time-frequency spectrum of the denoised signal from Figure 4(b) via the CPPCA-BIC-based noise reduction method. Similarly, the SNR conditions of the time-frequency spectra of the moving vehicle shown in Figures 5(a) and (b) are 35dB and 10dB, respectively. The denoised result from Figure 5(b) is shown in Figure 5(c).

We can see that the proposed noise reduction method works well not only in removing the noise component but also in reconstructing the target signal components including the torso- or bulk-Doppler and micro-Doppler components. Figure 6 further demonstrates the improvement on the features via our noise reduction method. As shown in Figure 6, compared with the noised features, the denoised features hardly vary with the SNR condition.

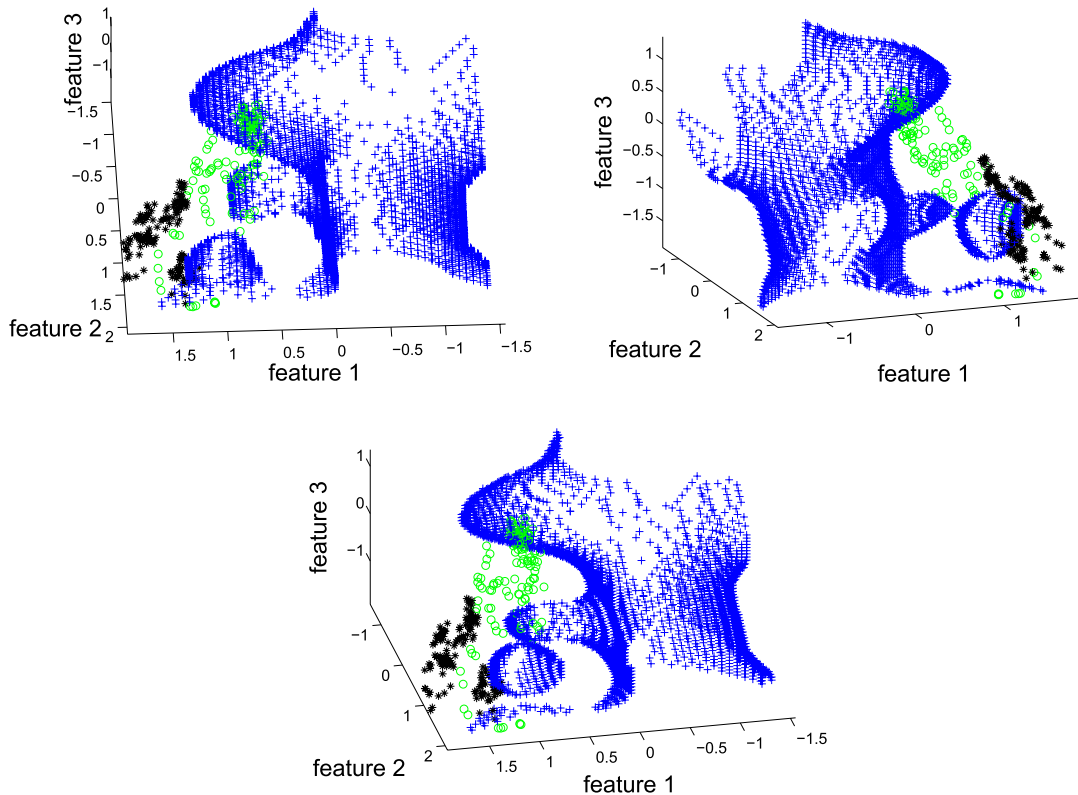


Fig. 7. Three-dimensional dispersion situations of the original time-frequency features from three different view-aspects for some measured data under $\text{SNR} = 10\text{dB}$. The green circle markers denote the feature data of a moving vehicle, the black star markers denote the feature data of a walking human and the blue plus markers denote the SVM classifier boundary learned via the feature data shown in Figure 2 under $\text{SNR} \approx 35\text{dB}$. The average classification accuracy rate of the feature data via the boundary in this figure is about 50%. The feature data shown in this figure are normalized with their mean and standard deviation.

B. Classification Results

In the classification experiment, we utilize the measured data to evaluate the classification performance of the noise-robust classification scheme introduced in Section III-B under the different test SNR conditions. Since the measured dataset are collected under high SNR condition ($\text{SNR} \approx 35\text{dB}$), a part of the measured data are used as the training data; while the rest of the measured data as the test data, which are added simulated noise corresponding to the different SNR cases. Here the noises in the inphase and quadrature echoes of targets are assumed to be Gaussian white noises. The SVM with Gaussian kernel is chosen as the classifier in our experiments. The parameter of Gaussian kernel is tuned in the range $[0, 5]$ for each classification method, and we select the best classification result from those of the different parameter settings for the following analysis and comparison.

Figure 7 shows the 3-dimensional dispersion situations of the original time-frequency features from the three different view-aspects for some measured data under $\text{SNR} = 10\text{dB}$, where the green circle markers denote the feature data of a moving vehicle, the black star markers denote the feature data of a walking human and the blue plus markers denote the SVM classifier boundary learned via the feature data shown in Figure 2 under $\text{SNR} \approx 35\text{dB}$. Obviously, the classification boundary learned under the high SNR condition cannot yield the good classification results for the features under the low SNR condition. The average classification accuracy rate of the

feature data via the boundary in Figure 7 is about 50%. The 3-dimensional dispersion situations of the denoised features, where the measured data under $\text{SNR} = 10\text{dB}$ are denoised via CPPCA-BIC before the feature extraction, are depicted in Figure 8. Comparing between Figures 2, 7 and 8, we can see that the dispersion situations of the denoised features are similar to those of the features under high SNR, therefore, the better classification performance can be guaranteed via the CPPCA-BIC-based denoising method under low SNR. The average classification accuracy rate of the feature data via the boundary in Figure 8 is about 85%.

In the following classification experiment, we compare the classification accuracies of the proposed noise-robust classification scheme with the noised features and the CLEAN-based noise reduction method. The CLEAN-based noise reduction method proposed in [7] uses the CLEAN algorithm to successively remove the large harmonic components in the noised echo until the residual energy smaller than the noise power, and the denoised data are obtained via reconstruction with the the large harmonic components. As shown in the experimental results in [7], the noise-robustness of classification can be improved via the CLEAN-based noise reduction method. Nevertheless, this method requires the prior information about noise power or SNR. If this prior information is not estimated accurately, the classification performance will decrease.

Figure 9 depicts the classification accuracies of the proposed 3-dimensional time-frequency feature versus SNR, generated via the four classification schemes, including the

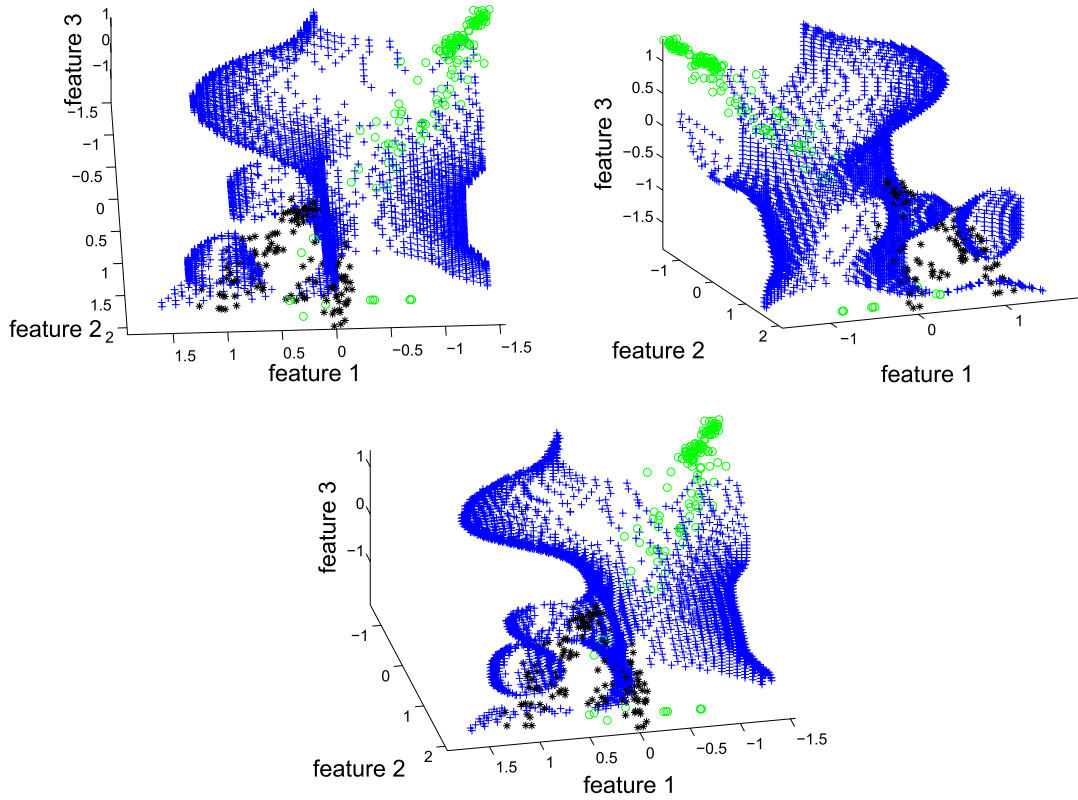


Fig. 8. Three-dimensional dispersion situation of the denoised time-frequency features from three different view-aspects. Here the measured data under $\text{SNR} = 10\text{dB}$ are denoised via CPPCA-BIC before the feature extraction. The green circle markers denote the feature data of a moving vehicle, the black star markers denote the feature data of a walking human and the blue plus markers denote the SVM classifier boundary learned via the feature data shown in Figure 2 under $\text{SNR} \approx 35\text{dB}$. The average classification accuracy rate of the feature data via the boundary in this figure is about 85%. The feature data shown in this figure are normalized with their mean and standard deviation.

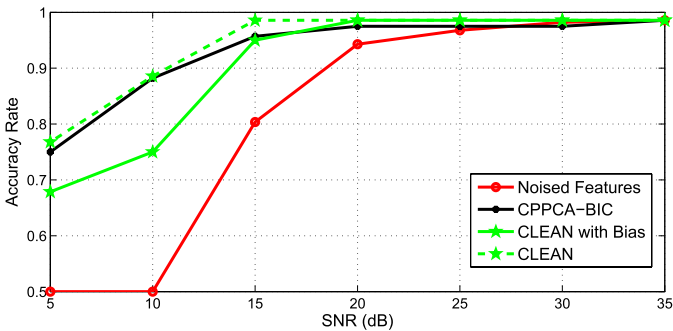


Fig. 9. Variation of the classification accuracy rates of the measured data with the test SNR via Noised Features, CPPCA-BIC, CLEAN with Bias and CLEAN. The training data are measured under high SNR condition ($\text{SNR} \approx 35\text{dB}$), while another part of measured data with simulated noises are used as the test data.

CPPCA-BIC-based noise-robust classification proposed in Section III-B, the CLEAN-based noise-robust classification with the accurate SNR information, the CLEAN-based noise-robust classification with the $\pm 3\text{dB}$ biased SNR information, and classification with noised features. For simplicity, the above four classification schemes are referred to as *CPPCA-BIC*, *CLEAN*, *CLEAN with Bias* and *Noised Features* respectively in Figure 9. When the $\text{SNR} \geq 25\text{dB}$, the influence of noise is not so significant, and the four methods yield

the similar classification accuracies; when $\text{SNR} \leq 25\text{dB}$, the three noise-robust methods outperform *Noised Features*; when $\text{SNR} < 15\text{dB}$, the biases on SNR affect the performance of *CLEAN*, and the classification accuracies of *CLEAN with Bias* are much lower than those of *CLEAN* and *CPPCA-BIC*; *CLEAN* is a little better than *CPPCA-BIC* only under $\text{SNR} \approx 15\text{dB}$, and they yield the similar classification results under the other SNR conditions. Therefore, *CPPCA-BIC* is more robust in the real application due to no SNR prior required. If we assume the classification rate threshold for the real application is 80%, *CPPCA-BIC* can work under the test condition of $\text{SNR} \geq 7\text{dB}$, while *Noised Features* requires $\text{SNR} \geq 15\text{dB}$. According to the radar equation, in the real application the 8dB advantage in SNR will bring an increase about 60% in the classification distance between radar and target.

V. CONCLUSION

In this paper, we proposed a noise-robust classification scheme to discriminate the moving vehicle and walking human via the time-frequency feature extracted from the micro-Doppler signatures of low resolution radar. The time-frequency spectrogram is used to extract a 3-dimensional feature vector for the two kinds of ground moving targets. In order to improve the classification performance under the low test SNR condition, a noise reduction method is developed based on

CPPCA model with BIC, where CPPCA model is derived for the complex-valued low-resolution radar echo and BIC is used to adaptively select the principal components for denoising. Compared with the existing CLEAN-based noise reduction method [7], the CPPCA-BIC-based method can work without SNR prior information. Experimental results based on measured data show that the proposed classification scheme can obtain good classification and denoising performances under the relatively low test SNR condition. The proposed method can also be applied to other classification problems based on micro-Doppler effect. In the real application, the SNR advantages of the proposed noise-robust classification scheme can effectively extend the classification distance between the target and radar.

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